

Sales Data Analysis Using Python

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INTRODUCTION

This project performs data analysis on a Sales dataset using Python libraries such as Pandas, NumPy, Matplotlib and Seaborn. The goal of the project is to analyze Sales of different items in different Cities.

INSTALL PACKAGE

- 1) pip install numpy
- 2) pip install pandas
- 3) pip install matplotlib
- 4) pip install seaborn

CODE IMPLEMENTATION

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Data preparation
df = pd.read_csv(r"C:\Users\DHRUV
SHAH\SalesAnalysis\Salesdata.csv")
print(df)
print("\n")
```

```
df['OrderDate'] = pd.to_datetime(df['OrderDate'])
print(df['OrderDate'])
print(type(df['OrderDate']))

df['Revenue'] = df['Price'] * df['Quantity']
print(df)
print("\n")

# Data analysis

Total_Revenue = df['Revenue'].sum()
print(Total_Revenue)
print("\n")

Revenue_by_PC =
df.groupby('ProductCategory')['Revenue'].sum()
print(Revenue_by_PC)
print("\n")

Top5_products =
df.nlargest(5, 'Revenue')[['ProductCategory', 'ProductNa
me']]
print(Top5_products)
print("\n")

Revenue_by_City = df.groupby('City')['Revenue'].sum()
print(Revenue_by_City)
print("\n")

mostUsed_Paymentmethod = df['PaymentMethod'].max()
print(mostUsed_Paymentmethod)

# Visualization
```

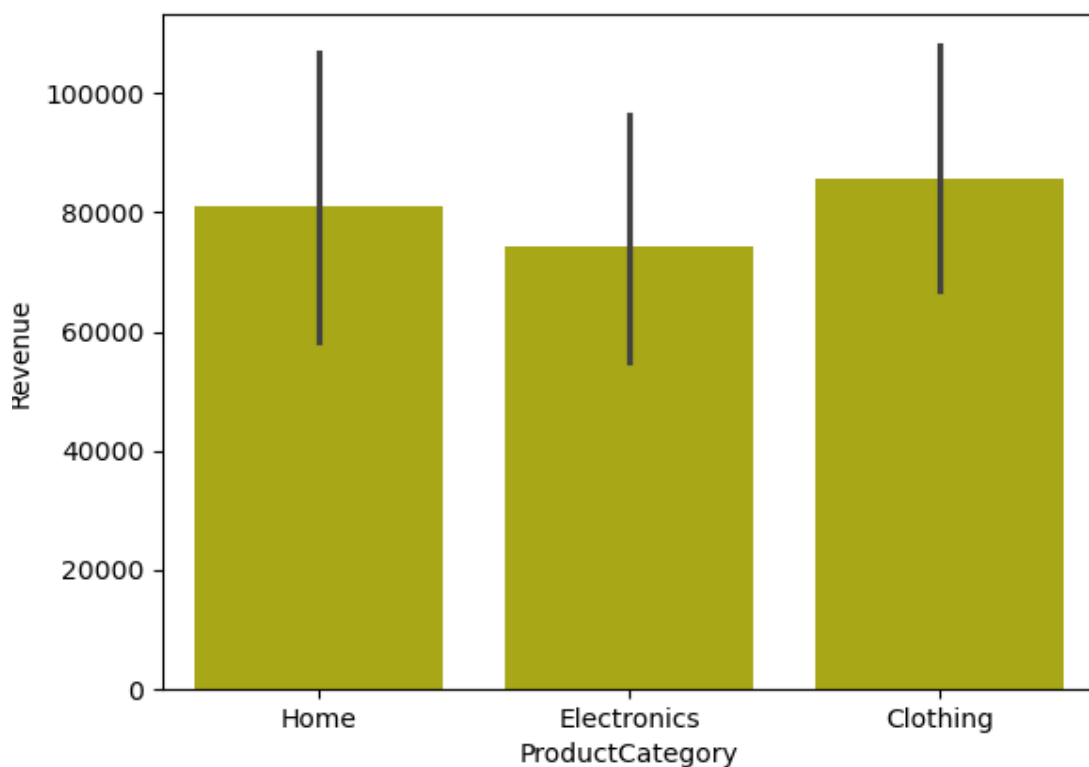
```
sns.barplot(x="ProductCategory", y="Revenue", data=df,  
color="y")  
plt.show()
```

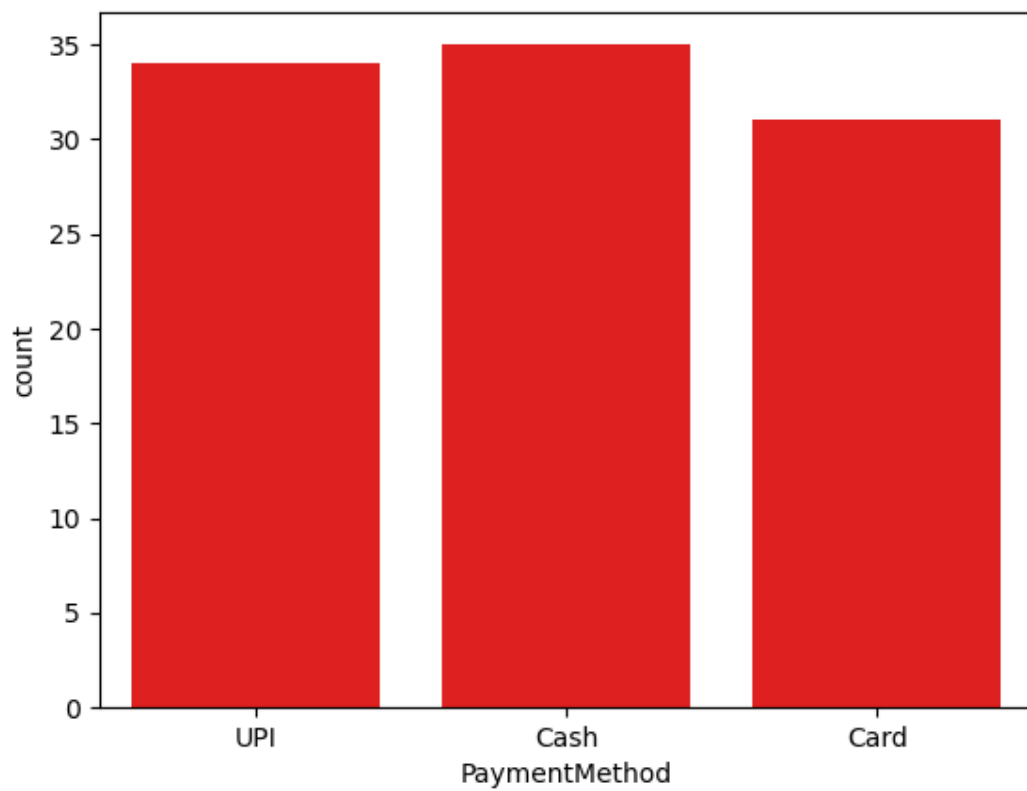
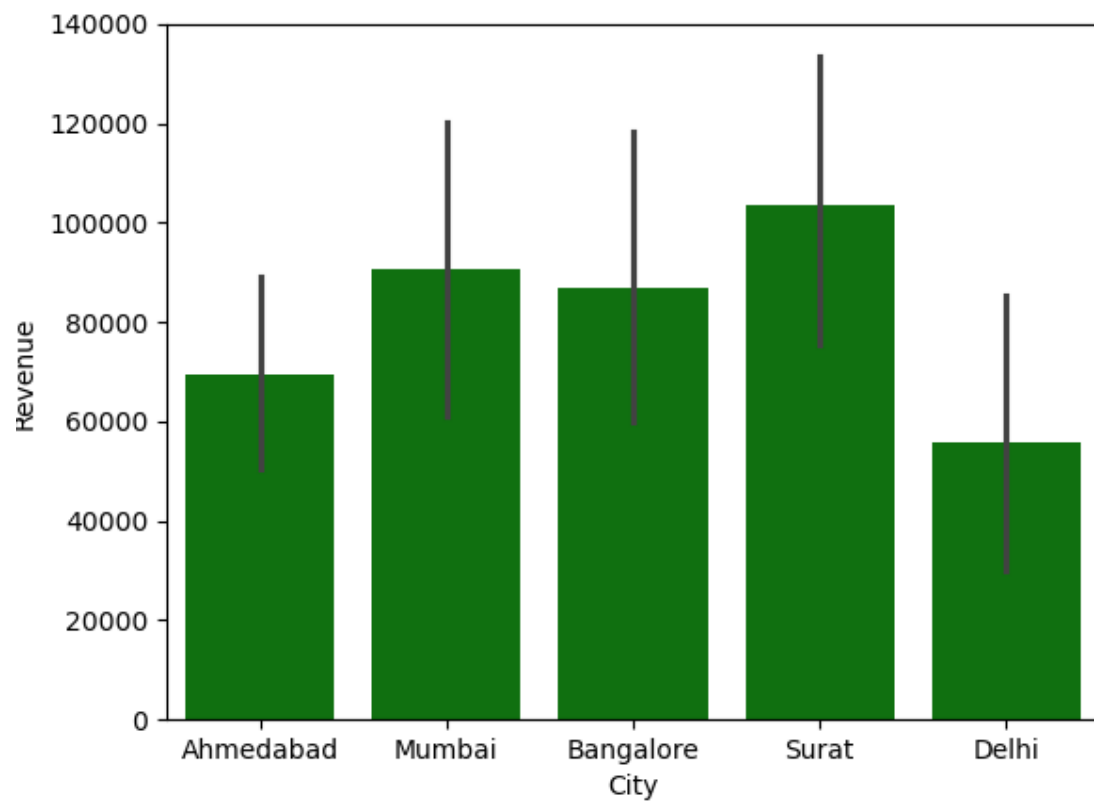
```
sns.barplot(x="City", y="Revenue", data=df, color="g")  
plt.show()
```

```
sns.countplot(x="PaymentMethod", data=df, color="r")  
plt.show()
```

""" INSIGHTS

- 1 Which category is most profitable? Clothing
 - 2 Which city performs best? Mumbai
 - 3 Which products generate highest revenue? Jackets
- """





OBSERVATIONS

- 1)The most profitable product category is Clothing with total revenue of 2,822,518.
- 2)The second highest revenue comes from Electronics.
- 3)The best-performing city is Mumbai, which generated 1,905,350 in revenue.
- 4)Bangalore performed second best with 1,738,707.